

Inline and Display Mathematics

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\Gamma_{1,fermions}[B] = \frac{1}{2} \int_0^\infty \frac{dT}{T} \int_{x(0)=x(T)} Dx(t) \exp(-\frac{1}{4} \int_0^T dt \dot{x}^2) Tr_L \Phi^{[1/2]}[\dot{x}] Tr_c P \exp(ig \int_0^T dt \dot{x} \cdot B)$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\zeta_{P\bar{K}}(s) = \zeta_{\bar{V}}(s) + \frac{\Gamma(s + \frac{1}{2})}{\sqrt{\pi} \Gamma(s)} \left[\frac{2}{3^{s+\frac{1}{2}}} {}_2F_1[\frac{1}{2}, s + \frac{1}{2}, \frac{3}{2}, -\frac{1}{3}] - \frac{1}{4s} \frac{1}{s} \right]$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\left(\frac{i\epsilon_{\mu\lambda\alpha} p_\lambda}{p^2} \right) \left(\frac{i\alpha\epsilon_{\alpha\sigma\beta} p_\sigma}{\sqrt{p^2}} \right) \left(\frac{i\epsilon_{\beta\kappa\nu} p_\kappa}{p^2} \right) = \frac{i\alpha\epsilon_{\mu\lambda\nu} p_\lambda}{(p^2)^{3/2}}.$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\left(\frac{1}{2} f' + e^f \frac{1}{12\sqrt{2}} \bar{g} (2e^{-\varphi} + e^{2\varphi}) \right) \gamma_0 \varepsilon_a = -\frac{1}{6} \sqrt{\frac{1}{2}} e^f e^\varphi \frac{1}{2\bar{g}} \gamma^{\nu\rho} R_{\nu\rho}{}^{ij} \gamma_{ij} \gamma_0 \varepsilon_a.$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\gamma_{g,9} : |D_{\text{inv}}\rangle \rightarrow \frac{1}{\sqrt{D}} \sum_{\alpha} \sum_{\beta, \gamma} c_{\alpha\beta} \bar{c}_{\gamma\alpha} |d_{\beta} \otimes \bar{d}_{\gamma}\rangle = \frac{1}{\sqrt{D}} \sum_{\alpha} |d_{\alpha} \otimes \bar{d}_{\alpha}\rangle = |D_{\text{inv}}\rangle,$$